

# John Doe

## Instrumentation & Control Engineer

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### PROFESSIONAL SUMMARY

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Instrumentation & Control Engineer with 8+ years of experience designing, configuring, and commissioning control systems for energy infrastructure. Proven expertise in DCS, PLC, HMI, and SCADA systems across complex industrial facilities. Strong background in instrumentation detailed design including schematics, I/O drawings, wiring diagrams, and control panel layouts. Experienced in leading engineering deliverables from conceptual design through commissioning, mentoring junior engineers, and ensuring compliance with industry codes and standards.

### CORE SKILLS

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- Process instrumentation design and specification
- P&ID development and interpretation
- Instrument index and specification sheets (ISA data sheets)
- Loop diagrams and wiring design
- Hook-up drawings and installation details
- DCS configuration (Honeywell Experion, Emerson DeltaV)
- PLC programming (Allen-Bradley, Siemens S7)
- SCADA systems and HMI design
- SIS design per IEC 61511 / ISA 84
- SIL verification and validation
- Instrument calibration and loop checking
- HART / Foundation Fieldbus / Profibus protocols
- Smart transmitter configuration (Emerson, Yokogawa, Endress+Hauser)
- Calibration management (Beamex CMX, Fluke tools)
- Project planning and execution
- Vendor technical evaluation and procurement
- FAT / SAT coordination
- Commissioning and start-up support
- Technical documentation and as-built records
- SmartPlant Instrumentation (SPI / INtools)
- AutoCAD / MicroStation
- MATLAB / Simulink
- SAP PM module
- Microsoft Office Suite
- Functional safety lifecycle management
- HAZOP and LOPA participation
- SIL assessment and classification
- Compliance: IEC 61511, IEC 61508, ISA 84, API 554, NEC/NFPA
- Gas facility instrumentation and control design
- Control and system functional narratives
- Telemetry and remote SCADA systems
- CSA codes and Canadian standards compliance
- MS-Project for project planning

### PROFESSIONAL EXPERIENCE

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#### Senior Instrumentation Engineer

Jan 2021 - Present

**Petronova Engineering** Houston, TX

- Lead instrumentation engineering for greenfield and brownfield projects valued at \$50M+
- Develop instrument specifications, data sheets, and hook-up drawings for pressure, temperature, level, and flow instruments
- Design and review P&IDs, loop diagrams, and cable schedules using SmartPlant Instrumentation
- Configure and commission DCS (Honeywell Experion) and SIS (Triconex) for refinery units
- Conduct SIL verification studies and prepare safety requirement specifications (SRS)
- Coordinate with multidisciplinary teams (process, electrical, piping) during FEED and detailed design
- Mentor junior engineers and review deliverables for quality assurance

## Instrumentation Engineer

Mar 2017 - Dec 2020

**Gulf Coast Process Solutions** Pasadena, TX

- Executed instrumentation detailed design for petrochemical and gas processing facilities
- Prepared instrument index, I/O lists, junction box layouts, and cable routing drawings
- Performed loop checking, calibration, and pre-commissioning of 2,000+ instrument loops
- Configured HART smart transmitters and control valves with digital positioners
- Supported DCS (Emerson DeltaV) point-to-point checkout and logic validation
- Participated in HAZOP reviews and implemented LOPA recommendations

## Junior Instrumentation Engineer

Jun 2014 - Feb 2017

**Meridian Industrial Services** Baton Rouge, LA

- Assisted in design and installation of instrumentation for power plant upgrades
- Created instrument hook-up drawings and typical installation details in AutoCAD
- Performed field calibration of pressure, temperature, level, and flow transmitters
- Maintained calibration records and updated instrument database in SPI/INtools
- Supported plant turnaround activities including instrument inspection and replacement

## EDUCATION

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### B.Sc. in Electrical Engineering (Instrumentation & Control)

2014

**Louisiana State University** Baton Rouge, LA

## CERTIFICATIONS

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- Certified Functional Safety Engineer (CFSE) - TUV Rheinland
- ISA Certified Control Systems Technician (CCST Level III)
- OSHA 30-Hour General Industry Safety

## PROFESSIONAL MEMBERSHIPS

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- International Society of Automation (ISA) - Senior Member
- Society of Petroleum Engineers (SPE)

## LANGUAGES

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- English - Native
- Spanish - Professional working proficiency